

Algebra 1
Unit 1
Lesson 7 Homework

Name _____
Date _____
Period _____

1. Write the expression $64^{\frac{3}{2}}$ in two different equivalent forms.

Expression 1	Expression 2

2. The value of $64^{\frac{3}{2}}$ is

- ☐ 16
- ☐ 42
- ☐ 83
- ☐ 512

For questions 3-5, write an equivalent radical expression and then find the value of the expression.

3. $(-216)^{\frac{2}{3}} = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

4. $-\left(\frac{32}{243}\right)^{\frac{4}{5}} = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

5. $(144)^{-\frac{1}{2}} = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

Evaluate each expression in questions 6-10 without a calculator.

6. $-(2^{-2})(81^{-\frac{1}{2}})$

7. $(32^{\frac{1}{5}})^3 \cdot 8^{-1}$

8. $-\left(\frac{16}{64}\right)^{-\frac{3}{2}} =$

9. $(-1^{10}) \cdot (-125^{\frac{1}{3}})$

10. $\left(\frac{7}{21}\right)^{-3} \cdot (3^{-2})$